



Föreläsning: 11

Datorer och spelkonsoler

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Innehåll



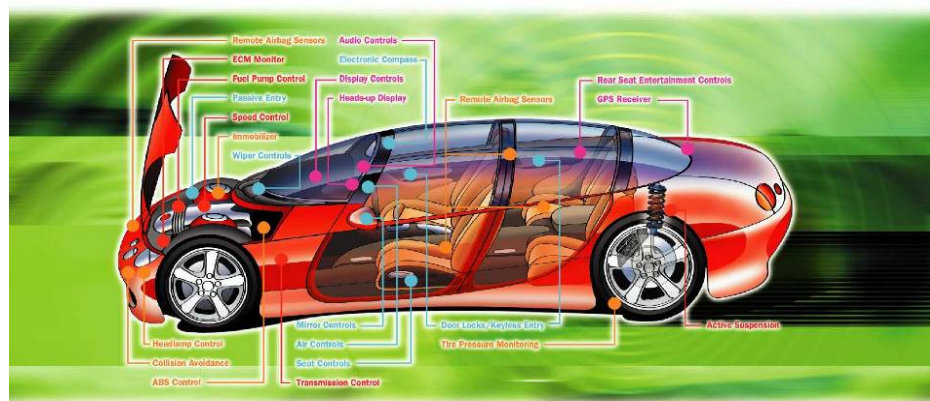
- Vad är "datorer" och var finns de?
- Lite historia
- Datortyper och uppbyggnad av en "typisk PC"
- Kort om spelkonsoler
- Något om speciella processorer

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Var finns "datorer" idag?



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Var finns "datorer" idag?



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Var finns "datorer" idag?

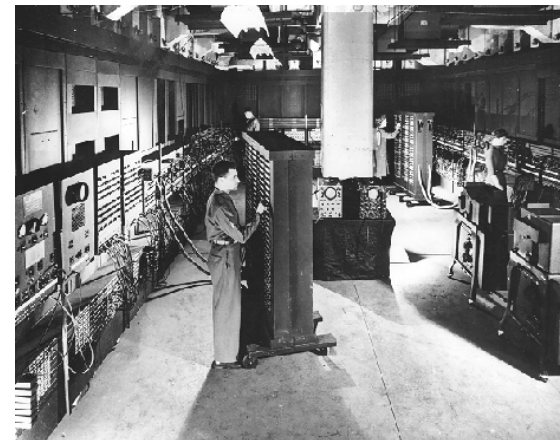


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Den "första" elektroniska datorn



ENIAC-1946

18 000 rör, 30 ton, 150m² ,140kW

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Den första elektroniska datorn



"I think there is a world market for maybe five computers."

-- Thomas Watson, chairman of IBM, 1943

"Computers in the future may weigh no more than 1.5 tons."

-- Popular Mechanics, forecasting the relentless march of science, 1949

"640K ought to be enough for anybody."

-- Bill Gates, 1981



ENIAC-1946

18 000 rör, 30 ton, 150m² ,140kW

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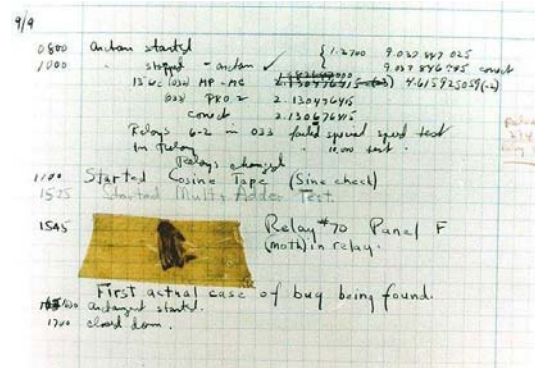
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Första buggen



At Harvard one August night in 1945, Grace Hopper and her associates were working on the "granddaddy" of modern computers, the Mark I. "Things were going badly; there was something wrong in one of the circuits of the long glass-enclosed computer," she said. "Finally, someone located the trouble spot and, using ordinary tweezers, removed the problem, a two-inch moth.

From then on, when anything went wrong with a computer, we said it had bugs in it."

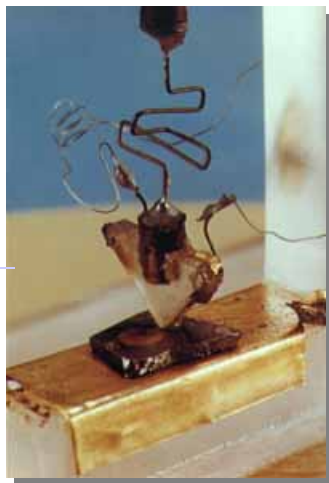


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Den första transistorn, 1947

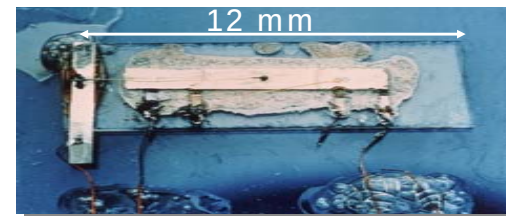


Shockley Brattain Bardeen

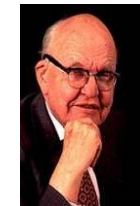


Nobelpriset 1956

Första integrerade kretsen, 1958



Kilbys IC: Phase-shift oscillator, 1.3MHz
5 komponenter varav 1 Transistor



Jack Kilby
Nobelpriset 2000

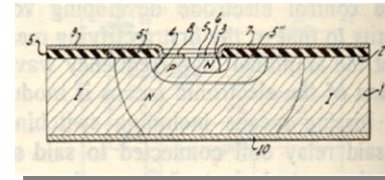


Bild i Noyce patentansökan



Robert Noyce
Co-founder of Intel

Moore's Lag

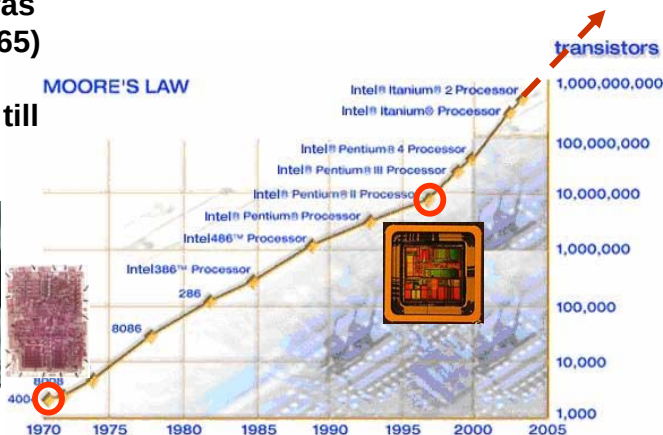


Antalet transistorer per chip dubblas var år. (1965)

Ändrar 1975 till vartannat år.



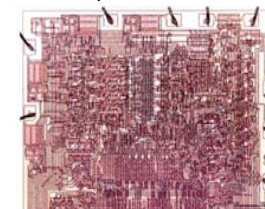
Gordon Moore
En av Intels grundare



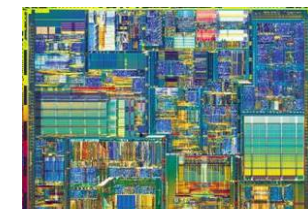
4004 till Pentium



Intel 4004 (1971)
2300 transistorer
10µm/108kHz



Intel Pentium 4 (2000)
42 millioner transistorer
0.18µm / 1.5GHz



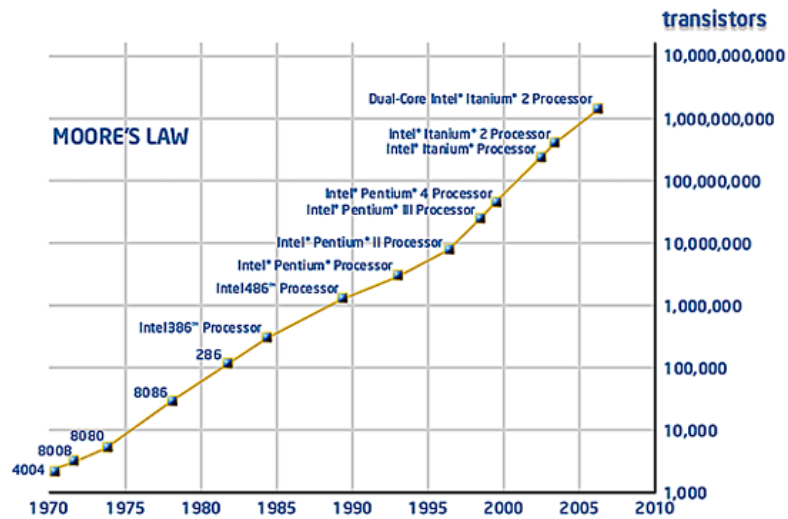
approx.

Om hastigheten för en bil hade ökat lika mycket som klockfrekvensen hade man kunnat köra från New York till San Fransisco på 13 sekunder!



18 000 gångerna fler transistorer på 29 år!

Moores lag: senaste notering



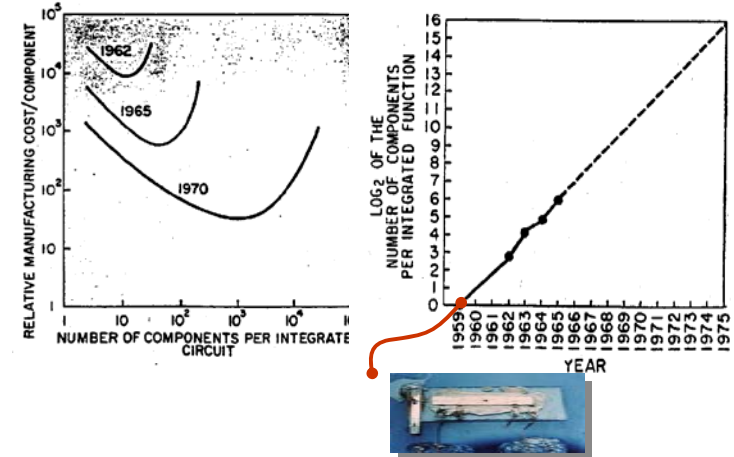
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Originalartikeln, 1965:

Cramming more components onto integrated circuits



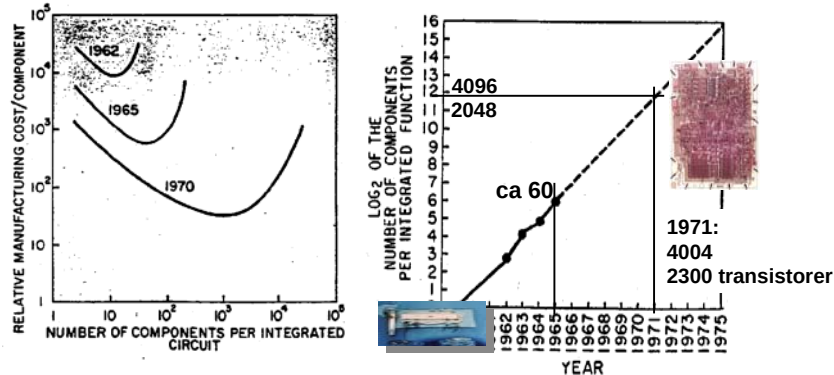
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Originalartikeln, 1965:

Cramming more components onto integrated circuits



"I was just trying to get across the idea that chips were going to get more complex and because of that the cost per transistor was going to drop dramatically."

Interview in 2000

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Originalartikeln, 1965:

Cramming more components onto integrated circuits



"Integrated circuits will lead to such wonders as **home computers** . . . and **personal portable communications equipment**"

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Och nu till de "riktiga" datorerna!

Mainframe and Super Computers



Mainframe Computers
Reliability and availability
Banks, airlines, government,....



IBM System
z9 109
1-54 processors
1740kg
18.3kW
512GB

Supercomputers

Designed for heavy computations



CRAY X1
8192 processors
2610kg
65kW
Liquid cooled



CRAY-2, 1987
"With an astonishing
2GBytes of main
memory"

Desktop, Laptops and PDAs



Apple-II, 1977
1 MHz, 4-48 kB of RAM



Apple Lisa, 1983
5 MHz, 1 MB of RAM



1979 by a Briton
340K byte bubble

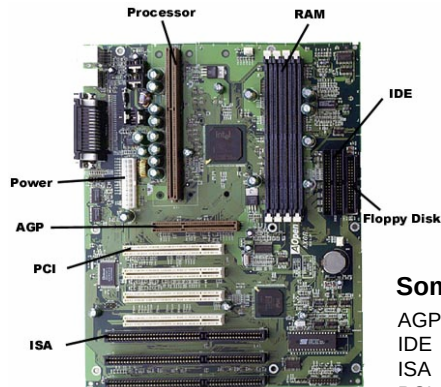


Osborne 1
1981



What is a "computer"?

Mother board



Some terminology

AGP	- Accelerated Graphics Port
IDE	- Integrated Drive Electronics
ISA	- Industry Standard Architecture
PCI	- Peripheral Component Interconnect
RAM	- Random Access Memory

Central processing unit (CPU)

- the "brain of the computer"



The CPU executes a program stored in a memory

ALU - Arithmetic/Logic Unit
 • addition, subtraction, ...
 • and, nand, or,...

Program and data have to be "fetched".
 CPU has "registers" for temporary storage.

Floating point unit

Memory operations

Written in

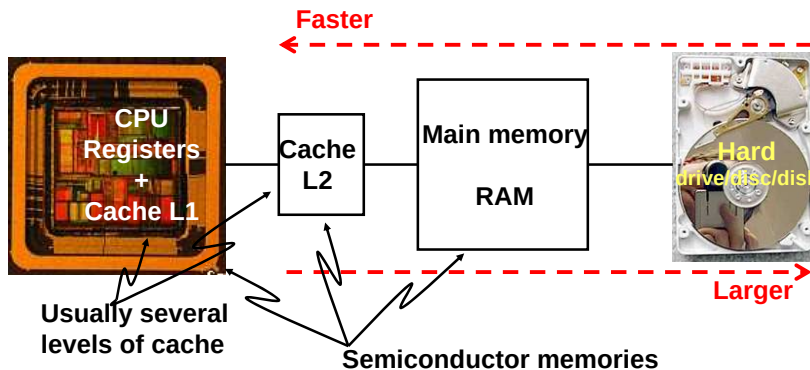
- High level, e.g. C, Java, etc
- Assembly code
- load, store, add,...

Transformed (compiled) into bit patterns

Memory Hierarchy



Larger memories become slower $\Rightarrow$
 Often several layers of memory hierarchy is used to have both large storage capabilities and fast memory acces



Memory Hierarchy, speed & size



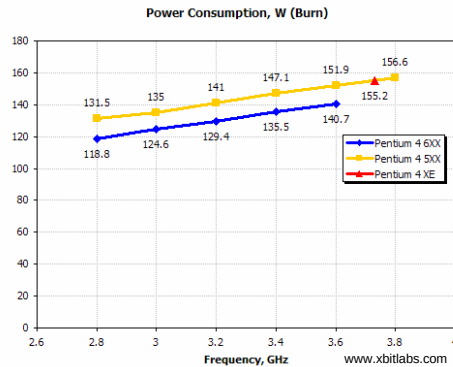
Cache Level 1: Full processor speed
 kBytes

Cache Level 2: "2-3 times" slower
 MBytes

Main/RAM: again "2-3 times" slower
 hundreds of MBytes $\Rightarrow$ GBytes

Hard drive: slower
 GBytes $\Rightarrow$ hundreds of GBytes

CPU power consumption



Pentium IV chip area (in 130 nm technology) 1.3 cm²

This gives about **100 W/cm²** that needs to be transported away (cooling)

Comparison: This little thing operates at about **10 W/cm²**.



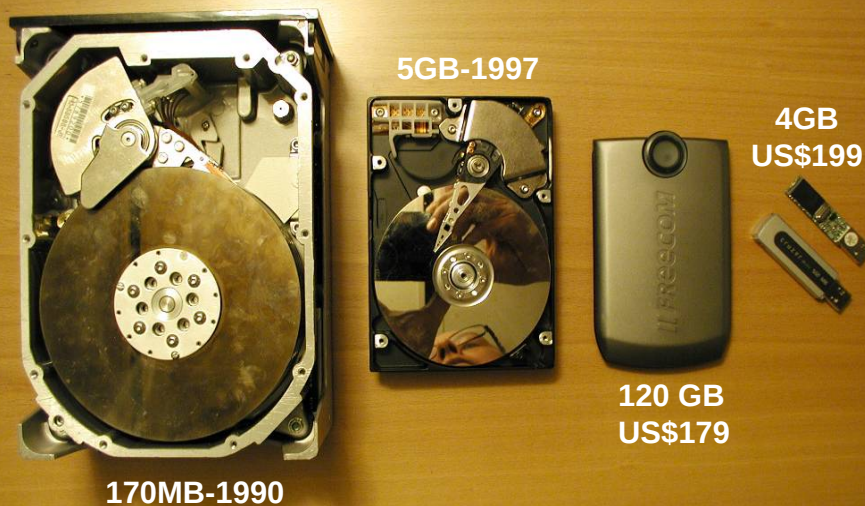
CPU power consumption



Some resort to **water cooling systems** for their processors!



Utvecklingen av massminnen



Mass storage



Interfaces

Advanced Technology Attachment (**ATA**)
aka Integrated Drive Electronics (**IDE**)
Serial ATA (**SATA**)
Small computer system interface (**SCSI**)
(Parallel Port, USB, FireWire ...)

Fixed: Hard drive

Removable:

Semiconductor – memory stick, Flash cards, ...

Magnetic – Hard drive, Tape, Floppy disk, ZipDrive

Optical - CD / CD-R / CD-RW, DVD/ DVD-R /
DVD+R / DVD-RW / DVD+RW

Basic input/output system (BIOS)



The BIOS makes sure all the other chips, hard drives, ports and CPU have a basic set of communication possibilities at startup and can function together.

A program stored on a Flash Memory, so that it can be upgraded if necessary.



Operating system (OS)



Manages the hardware and software resources.

Provides a stable, consistent way for applications (programs) to deal with the hardware without having to know all the details of the hardware.

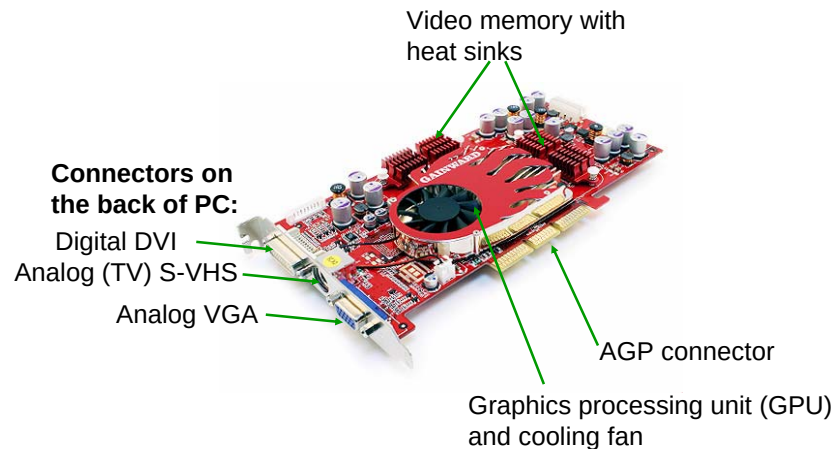
Different types:

- Real-time
- Single user/single task
- Single user/multiple tasks
- Multiple user/multiple tasks

Examples:

- MS-DOS
- MS-Windows
- Novell Netware
- Mac OS
- Unix
- Linux

Graphics adapter

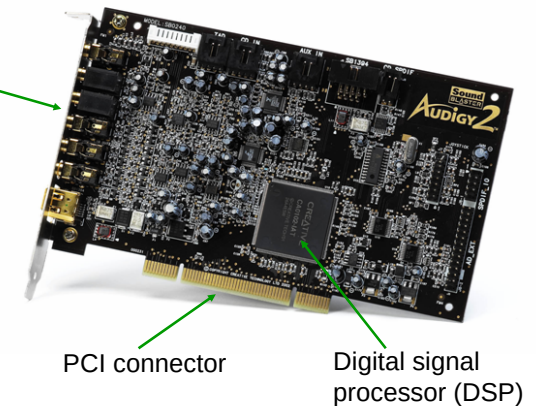


Sound adapter

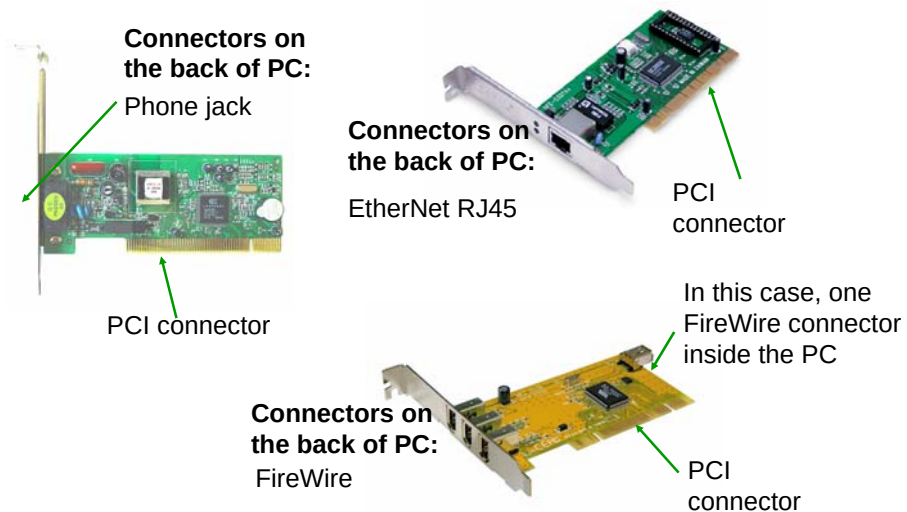


Connectors on the back of PC:

Analog and digital in- and output



PhoneModem / Network / FireWire



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Power supply



Converts 110/230 V AC to +/- 12 V, 5 V, 3.3 V, ... DC.

One output contact for the motherboard and several others for other units (disk drives, cd-rom drives).

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Ports and connectors



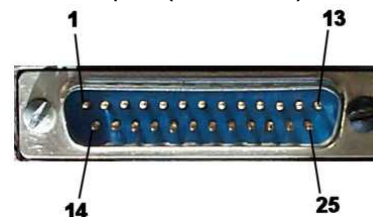
Parallel and serial ports



PS/2

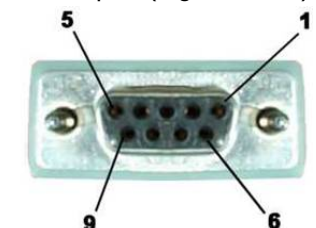


Parallel port (IEEE 1284)



8 bits in parallel
standard: 50-100kbytes/s
enhanced: 0.5-2Mbytes/s

Serial port (e.g. RS-232)



1 bit at a time
standard: 115kbit/s
enhanced: 460kbit/s

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EtherNet (IEEE 802.3)



Most common, RJ45
(Registered Jack)



"Older" versions, BNC
(British Naval Connector)



Universal Serial Bus (USB)



USB 1.1 - 1.5 Mb/s and 12 Mb/s

USB 2.0 - 480 Mb/s

Applications:

Mouse, keyboard, tablet,
memory stick, digital camera,
external CD-ROM,
video camera (USB 2.0),
external hard disks (USB 2.0) etc.

Type A



Type B



FireWire (IEEE 1394)



FireWire 400: IEEE 1394 – 400 Mb/s



FireWire 800: IEEE 1394b S800 bilingual – 800 Mb/s



Applications:

External hard disks, video
cameras, etc.

Special processors



Digital Signal Processors (DSPs)



Used where signal processing tasks are important, e.g., in sound cards, mobile phones,



Similar to ordinary CPUs in many ways, but

very good at performing series of multiply/add operations which are common in filtering applications

have high performance in/out capabilities enabling them to handle rapid streams of digital signals

A common requirement is real-time performance!

Graphics Processing Units (GPUs)

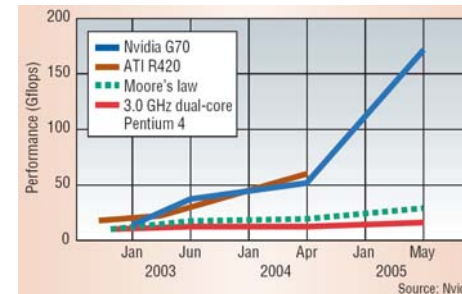


Used where image processing tasks are important, e.g., in graphics cards, game consoles etc.

Similar to ordinary CPUs in certain ways, but usually "not as programmeable"

have many parallel processing units (vector operations)

very good at rendering graphics primitives (such as polygons)



This development makes them very interesting for specialized number crunching applications.

Video Game Consoles



Why video game consoles when we have PCs?



- Standardized contents (hardware/software)
 - Can be produced in large numbers at low cost.
 - Makes them easy to handle for users, no problems with wrong versions of drivers or operating systems
 - Game developers know exactly the features of the platform they are developing games for (can make program code very efficient)
- Only functions essential to gaming are implemented, making their "operating systems" much simpler and faster.
- Intended for entertainment and can therefore be connected directly to a standard TV.

Sony PlayStation 2



- Processor: 128-bit "Emotion Engine"
 - 300 MHz
 - Floating point unit (FPU) co-processor
 - Maximum bus transfer rate of 3.2 GB per second
 - Includes current PlayStation CPU core
- Graphics: "Graphics Synthesizer"
 - 150 MHz
 - Embedded cache
 - 4 MB VRAM
 - **75 million polygons per second**
- Audio: SPU2 (+CPU), 48 channels, 44.1- or 48-kHz sampling rate, 2 MB memory
- RAM: 32 MB RDRAM
- Game medium: Proprietary 4.7-GB DVD and original PlayStation CDs
- Drive bay (for hard disk or network interface)
- Controller: Two controller ports, "Dual Shock 2" analog controller
- Other features:
 - Two memory card slots
 - Optical digital output
 - Two USB ports
 - FireWire port
 - Support for audio CDs and DVD-Video



from www.howstuffworks.com

Nintendo GameCube



- Processor: "Gekko" IBM Power PC microprocessor
 - 485 MHz
 - Cache:
 - level 1: 32 KB Instruction and 32 KB Data
 - level 2: 256 KB
 - 32-bit address, 64-bit data bus
 - Maximum bus transfer rate of 2.6 GB per second
 - 0.18 micron copper interconnects
- Graphics: "Flipper" ATI graphics chip
 - 162 MHz
 - 1 MB embedded texture cache
 - 3 MB Mosys 1T-SRAM (This static RAM uses a single transistor per cell, like DRAM.)
 - **Approximately 12 million polygons per second**
- Audio: Special 16-bit digital signal processor, 64 channels, 48-kHz sampling rate
- RAM: 40 MB (24 MB 1T-SRAM, 16 MB of 100-MHz DRAM)
- Game medium: Proprietary 1.5-GB optical disc
- Controller: Four game controller ports, Wavebird wireless game controller
- Other features:
 - Handle for carrying
 - Two slots for 4-MB Digicard Flash memory cards or a 64-MB SD-Digicard adapter
 - High-speed parallel port
 - Two high-speed serial ports
 - Analog and digital audio-video outputs



from www.howstuffworks.com

Microsoft Xbox



- Processor: Modified Intel Pentium III
 - 733 MHz
 - Maximum bus transfer rate of 6.4 GB per second
- Graphics: Custom nVidia 3-D graphics chip
 - 250 MHz
 - **Approximately 125 million polygons per second**
- Audio: Custom 3-D audio processor
- RAM: 64 MB (Xbox has a unified memory architecture -- the memory can be allocated to graphics, audio, textures or any other function as needed.)
- Game medium: Proprietary 4.7-GB DVD
- Modem/network: Media communications processor (MCP), 10/100-Mbps Ethernet, broadband enabled, 56K modem (optional)
- Controller: Four game controller ports
- Other features:
 - 8-GB built-in hard drive
 - 5X DVD drive with movie playback
 - 8-MB removable memory card
 - Expansion port



from www.howstuffworks.com

Nästa föreläsning



- Kringutrustning till datorer och spel
- "Inmatning"
 - Mus
 - Tangentbord
 - etc ...
- Utmatning
 - Skrivare
 - "force feedback"
 - etc ...